



Sinusoidal Gust Response of RC Propellers at Different Incidence Angles

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We investigate propeller performance under steady and sinusoidally time-varying freestream velocity in an open-jet wind tunnel. Changes in thrust, power, pitching moment, and rolling moment observed with increase in propeller incidence angle under steady freestream was then used to predict response under sinusoidal time-varying freestream. At both low and high incidence angles, the propeller response agrees with the prediction, suggesting linear superposition and quasi-steady dynamics for reduced frequencies of up to 0.0796.

I. Nomenclature

C_T	=	Propeller Thrust coefficient; $C_T = \frac{T}{\rho n^2 D^4}$	Q	=	Torque (Nm)
C_P	=	Power Coefficient; $C_P = 2\pi C_Q = \frac{P}{\rho n^3 D^5}$	R	=	Propeller Radius (m)
C_Q	=	Propeller Torque Coefficient; $C_Q = \frac{Q}{\rho n^2 D^5}$	T	=	Thrust, (N)
D	=	Propeller Diameter, (m)	V_∞	=	Freestream Velocity (m/s)
d	=	Distance, (m)	ρ	=	Air Density, (kg/m^3)
J	=	Propeller Advance Ratio	β	=	Shuttering System Louver Angle (deg)
J_z	=	Normalized Propeller Advance Ratio	σ	=	Propeller Solidity; $\sigma = \frac{s}{0.25\pi D^2}$
n	=	Propeller RPS	θ	=	Rotor Incidence Angle (deg)
P	=	Power, (W)	ω	=	Shuttering System Frequency (Hz)
p	=	Pressure, (Pa)	ω_k	=	Normalized Shuttering System Frequency
γ	=	Propeller Pitch, (in)	$\Delta\lambda$	=	Phase Lag (rad)
			η	=	Propeller Efficiency; $\eta = \frac{C_T J}{C_P}$

II. Introduction

The increasing usage of propeller-driven unmanned aerial vehicles (UAVs, or drone) broadens the classical helicopter problem of edgewise flight, where the rotors producing the vehicle lifting-force are subjected to a free-stream nearly parallel to the rotor-plane. Drones may plausibly encounter a continuum of incidence angles between the rotor(s) and the freestream, from purely edgewise flight to directly head-on, in the sense of a

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conventional airplane. Large incidence angles are to be expected when operating at small lift to drag ratio, for example with a light-weight drone flying near its top speed. Gust encounters or aggressive maneuver add the complication of time-varying relative freestream, where not only is there a variation in blade sectional cumulative speed magnitude from advancing/retreating blades, but from change in the oncoming free-stream itself.

The large literature on helicopter rotor unsteady response is motivated by wake interaction between blades [1-3], and by flapping, fluttering or other issues of rotor dynamics. Motivated by the growing need for the use of multi-rotor unmanned aerial vehicle (UAV) for business, commercial, and military purposes, recent study of drone-type propellers include ground effect [4], ceiling effect, and unsteady environments. Unsteady conditions cause delay in response to control input [5] as well as changes the power consumption. Power and thrust limitations due to the UAV sizing, to counter such unsteady effects by feedback control, also add delay [6-7]. Unsteady effects causing by the multi rotor tip vortex interaction and wake from the drone structure and external payload also affect the performance of the multi-rotor done, especially in higher speed forward flight [8].

Time-varying freestream velocity, or wind gust, has also been studied on helicopter rotors theoretically and via CFD [9-12]. Most of these theories are an extension of Theodorsen's unsteady airfoil theory [13]. Since the drone rotor is rigid [14], concerns of flapping/fluttering of helicopter blades [1] do not directly apply. While both helicopter and propeller blades experience azimuth-varying loading in nonzero relative freestream, fixed pitch propellers lack the cyclic control or passive cyclic degrees of freedom.

Propeller performance in steady freestream at different rotor incidence angles on variable [15] and fixed-pitch [16] propellers has been studied previously. Simmons and Hatke [16] investigated a 16-inch diameter, 8-inch pitch, three-bladed foldable propeller for eVTOL UAVs in steady freestream at an incidence angle, θ , between 0 and 180 degrees. As comports with the usual propeller theory, both C_{Tz} and C_{Qz} decrease with increasing J at small incidence angle. But the C_{Tz} and C_{Qz} vs. J curve flattens out at higher incidence angle. C_{Tz} and C_{Qz} are nearly independent of J at $\theta > 60^\circ$, and even increase with J upon approaching true edgewise flight.

Both pitching moment coefficient, C_{Qx} , and rolling moment coefficient, C_{Qy} , increase with increasing propeller incidence angle. Both are due to the uneven blade loading between the advancing and retreating blade. C_{Qx} increases with J , while C_{Qy} remains relatively constant. Since theoretical models tend to lose accuracy at higher incidence angles, the objective in the current study is to compare the simplest models with experiment data, and to combine the conditions of large rotor incidence and unsteady freestream.

III. Experimental Setup

A) Data Acquisition

Experiments were conducted at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open jet test-section configuration. The wind tunnel has an inlet contraction ratio of 16:1, a 762 mm x 762 mm testing section inlet, and a 1370 mm x 1370 mm collector. The freestream range is from 3 m/s to 37 m/s, resulting in advance ratio, J , between 0 and 0.7, with a turbulence intensity of 0.1% measured by a hot wire anemometer at 15 m/s.

A sketch of the experimental setup is shown in Figure 1. The propeller and motor are mounted at the centerline of the test section, 610 mm from the inlet, and attached to an ATI Industrial Automation Mini-40 six-component force/torque transducer [17]. Thrust is the axial-force along the z-axis of the sensor, and normal force is along the y-axis. The motor torque is the moment about the z-axis, while the propeller pitching moment is the

torque on the x-axis, and the propeller rolling moment is the y-axis torque. The balance is bolted to an aluminum-extrusion frame, which has two pivot points, to change the propeller incidence angle. The balance has a maximum of 40N on its x and y-axis, 120N on its z-axis, and maximum torque of 2 Nm for all axes. The force and torque data sampling rate was 1000 Hz, with two runs to improve data repeatability for each data point. In an effort to minimize sensor drift, tare values are taken before and after each test, and then the average of the two tares is subtracted from the respective thrust and torque readings. A band-pass filter is then applied to filter out any undesired frequency in the signal, including the rotor RPS and its harmonics, and the aluminum frame natural frequency (~ 65 Hz).

A Dantec Dynamics [18] P5516 Probe CTA hotwire is mounted more than 2 propeller diameters upstream the propeller, at the centerline of the tunnel plenum, to measure the instantaneous free-stream u-component of velocity. The hotwire is connected to a Dantec Dynamics 54T30 Mini-CTA and a National Instruments [19] USB-6259 DAQ. The hotwire is calibrated from 0-20 m/s, measured by a TSI 5825 Micro anemometer. A Thor Labs [20] 5 mW laser and a DET10A photodiode are used to measure the propeller rotational speed, also connected to the NI-USB-6259 DAQ. Both hotwire and photodiode are sampled at 5,000 Hz.

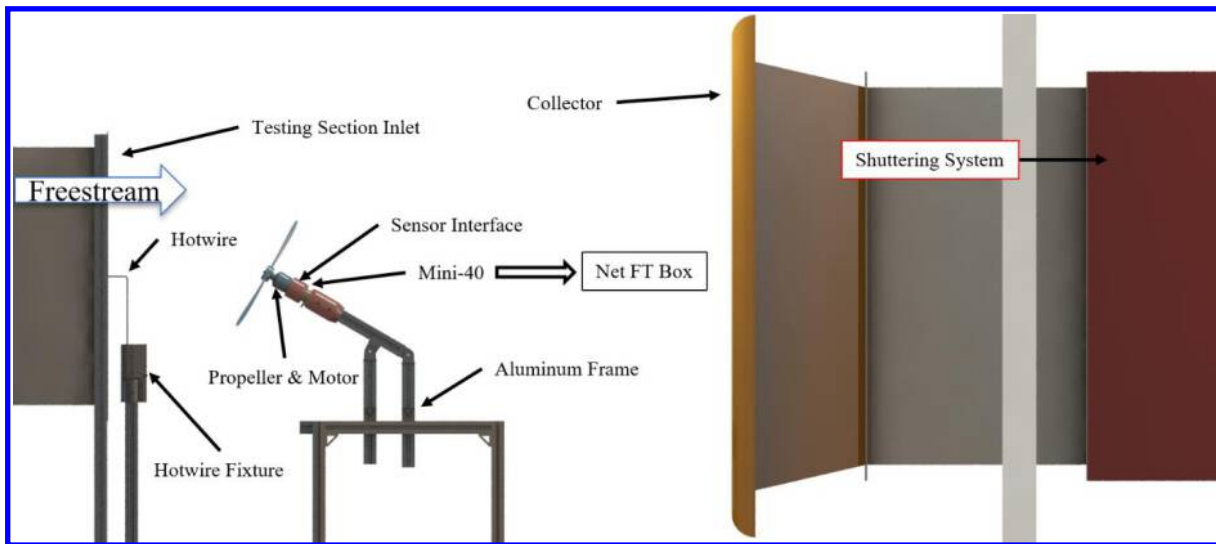


Figure 1. Schematic of the experimental setup

The Aerotech shuttering system is installed downstream of the test section, between the collector and diffuser. The system consists of a set of primary counter-rotating flat-plate vanes mounted in a plane normal to the flow, and a pair of side louver vane sets mounted in the sidewalls of the diffuser. The primary louvers operate over a louver angle β , range of 41° from full open flow to closed flow, while the secondary vanes are intended to minimize large scale flow separation. An example of the primary louvers fully open and closed is shown in Figure 2. The shuttering system is operated by a National Instruments PCI-7340 motion control card, controlled by LabVIEW at a system frequency of 125 Hz. The shuttering system linear actuator encoder position, F/T transducer, hotwire, and photodiode signal are all collected by LabVIEW on the same timestamp.

B) Test Matrix

All propellers used in the experiment are from the “thin electric” APC series, selected because of abundant evidence in the literature for their relatively high efficiency and good geometric repeatability across a broad range of diameters [21]. All propellers were driven by an E-Flite Power 60-400kV brushless out-runner electric motor. The propeller property and test matrix are shown below in Table 1. Two propellers were used. The 11x7 propeller

has a moderate pitch to diameter ratio, γ/D , while the 14x14 has a high γ/D , where blade stall would occur at small J [4]. The peak Reynolds number was calculated at the 70% propeller radius at peak rotational speed. 3 different incidence angles were chosen for the study between 0 and 90-degree, where the lower θ cases apply to a traditional fixed-wing UAV, higher θ to a quadcopter or eVTOL UAV, and the highest angle to a helicopter in its usual edgewise flight.

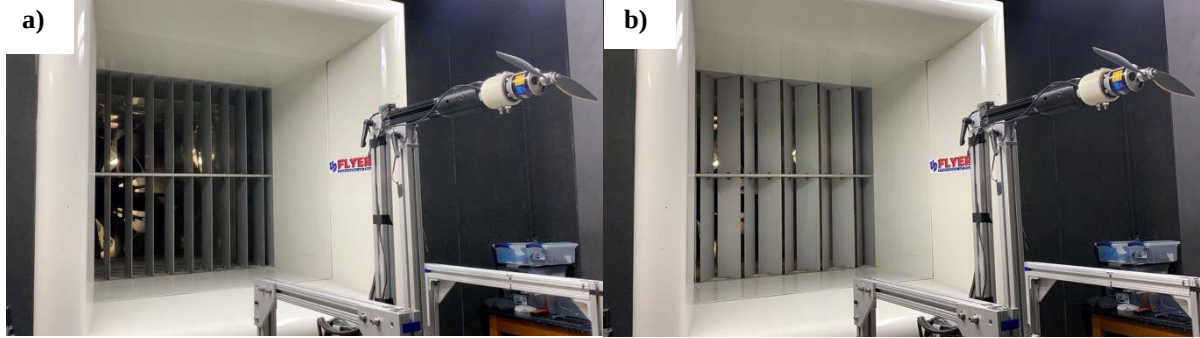


Figure 2. Shuttering system primary louvers a) fully open and b) fully closed

Table 1. Test matrix

Propeller	Rotational Speed per Second	Peak Reynolds Number (x1000)	Advance Ratio	Incidence Angle (Deg)	Shuttering System Frequency (Hz)	Freestream Velocity (m/s)	Normalized Shuttering System Frequency (10^{-2})
14x14	40~80	120	0 ~ 0.7	0, 45, 90	0.1, 0.4, 1.25, 1.75, 2.25	0~20 (Steady)	0.33, 1.47, 5.15, 7.13, 7.96
11x7	50~120	90				3~16 (Time Varying)	0.26, 1.16, 4.05, 5.60, 6.25

Two different methods are used to vary J . For the steady freestream velocity cases, where the shuttering system is not operating, the propeller rotational speed is kept constant, while the freestream velocity is put at different settings. For the time-varying freestream velocity cases, where the shuttering system is operating, the wind tunnel fan rotational speed is kept constant, but the propeller rotation speed is varied to vary advance ratio.

The shuttering system is operated at a frequency of 0.1 and 0.4 Hz (lower range), and 1.75 and 2.25 Hz (medium-high range), under wind tunnel fan speeds of 400 RPM. Shuttering system frequency is normalized by treating the propeller disk as a bluff body, of characteristic diameter D :

$$\omega_k = \frac{D}{V_\infty} \omega \quad (1)$$

Where $\overline{V_\infty}$ is the mean freestream velocity when shuttering system is operating, and ω is the shuttering system frequency. The resulting normalized shuttering frequency is also shown in Table 1.

IV. Shuttering System Characterization

In our previous study [22], it was identified that the open jet wind tunnel freestream velocity response to shutter-motion, is non-linear. With a sinusoidal shuttering system louver rotation input, the freestream velocity

response resembles a $\sin^3(\pi ft)$ profile at lower shutter frequency, and a $\sin^2(\pi ft)$ profile at higher shutter frequency. Several causes of the non-linearity are identified: expansion of the open jet in the test section; change of wind tunnel fan rotational speed due to pressure loss at higher louver angle; and additional inflow through secondary vanes into the diffuser. Here, a methodology will be discussed for generating a properly sinusoidal freestream velocity history.

A) Steady-State Response

The steady response of the test section freestream velocity under fixed louver angle, β , at difference tunnel fan RPM is shown in Figure 3. The relationship between the freestream velocity and β is non-linear; however, a linear portion occurs for $\beta \geq 20^\circ$. This indicates the possibility of limiting the louver angle to create a linear freestream velocity response to shuttering system input. However, simply limiting the louver angle does not provide a sinusoidal freestream velocity under a sinusoidal β input.

B) System Identification

The Matlab system identification toolbox was used to obtain an input β variation which would result in a sinusoidal velocity variation in the test section. Several models were used to model the system and the Nonlinear-ARX model [23] was found to be the most accurate. It is an extension of the linear ARX model, which includes a linear and a non-linear function to give a model output with a fixed offset. For the number of past input and output terms which are used to define the linear regression of the model, it is found out to be less or equal to 2 for all shutter frequency models.

Matlab Simulink is then used to build a ‘simulated’ closed-loop PID control system of the shuttering system. The PID controller is tuned with a relatively short response time, and thus, a high acceleration. The target profile for the control system is a sinusoidal freestream velocity with the same shuttering system frequency and freestream velocity magnitude as that for the model estimation. The controlled louver position based on the PID controller in the simulation is recorded and then applied to the ‘real-world’ shuttering system input with open-loop control.

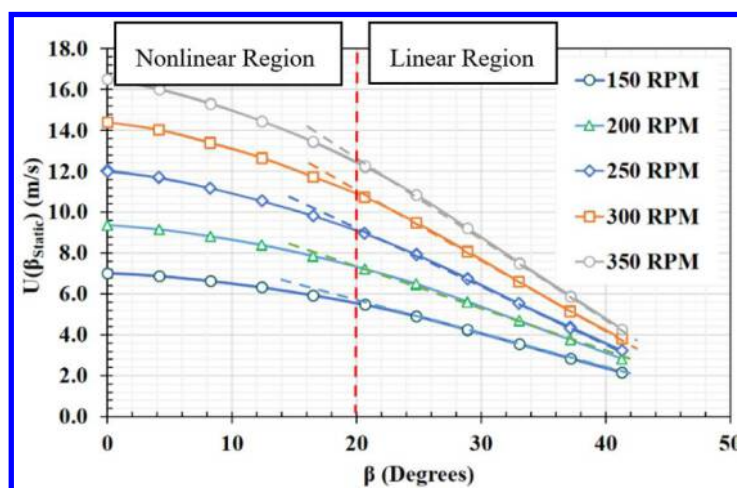


Figure 3. Test section mean flow speed vs. louver-angle, for a range of tunnel fan RPMs

Based on the steady-state wind tunnel response in Figure 3, the target freestream velocity magnitude is then limited to create a louver control profile within $20^\circ \leq \beta \leq 41^\circ$. The resulting normalized louver input profile, with ‘0’ representing the louver fully open (0°) and ‘1’ representing the louver fully closed (41°), along with the measured wind tunnel response, is shown in Figure 4 for 0.1Hz louver operational frequency, at regular/wider louver rotation angle and limited/narrower louver rotation angle. The measured freestream velocity is then

compared with a sinusoidal model, $y = f(\sin(kt))$. The resulting freestream matches the sinusoidal model better under limited louver rotation angle (where the $\beta - U_{\beta(static)}$ relationship is linear). The drawback of limiting the range of louver motion is the comparatively lower magnitude of velocity range that can be obtained as shown in Figure 4b. Similar trends were observed for all louver frequencies under 3 Hz.

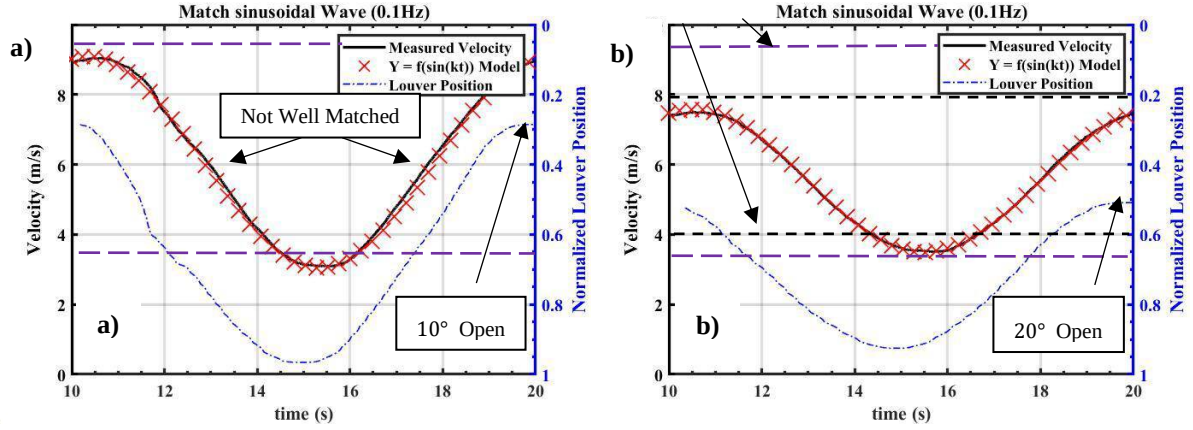


Figure 4. Measured freestream velocity vs. sinusoidal model at 0.1Hz louver frequency for a) full and b) limited louver rotation angle

Since the wind tunnel response varies under different shuttering system frequencies, the system identification is repeated for each louver frequency. The louver rotation angle and input shape vary for different louver frequency to produce a sinusoidal freestream. Figure 5 shows the louver input and the resulting freestream velocity compared with the sinusoidal model at 1.75Hz and 2.25 Hz louver frequency.

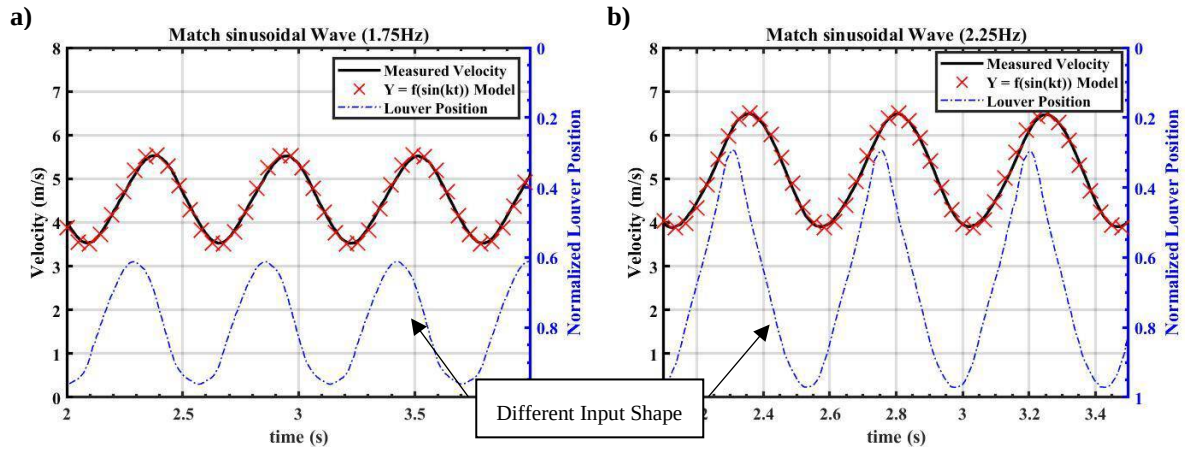


Figure 5. Measured freestream velocity compared with sinusoidal model for (a) 1.75 Hz louver frequency and b) 2.25Hz louver frequency

Figure 6 shows the mean, lower and upper bound of freestream vs. shuttering system frequency at fan RPM. The mean velocity, and the upper and the lower bound of the freestream velocity at different shuttering system frequency under different wind tunnel fan RPM after applying the methods discussed previously. At the same tunnel fan RPM, the mean value and the variation in freestream decrease with increment of ω at $0 < \omega < 1$ Hz, which was also seen in our previous study [22]. The mean value and variation then hold relatively constant at higher ω , until $\omega = 2.25$, where the mean freestream velocity increases slightly. With increase of the tunnel fan RPM, the freestream velocity increases accordingly.

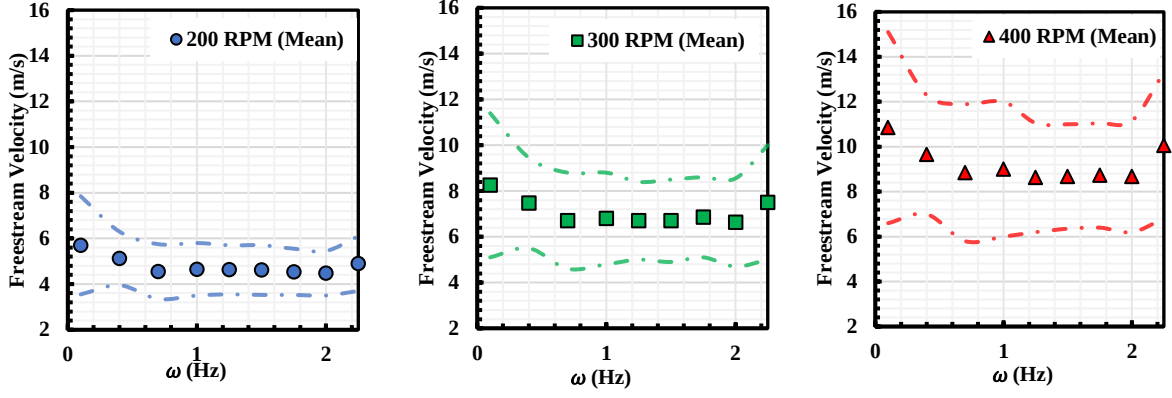


Figure 6. Mean, lower and upper bound of freestream vs. shuttering system frequency at fan RPM of 200, 300 and 400

V. Results

A) Propeller Performance in Steady Freestream

Each propeller is tested in steady freestream between 0 and 90-degree incidence angle, θ . At least three different rotational speeds are tested for each propeller to account for Reynolds number effects. To vary J , the propeller rotational speed was fixed while the tunnel freestream velocity was changed.

The axial thrust coefficient, C_{Tz} and power coefficient, C_p , are shown for the APC 11x7 propeller at 100 RPS in Figure 7. Results are consistent with [15] and [16]; coefficients decrease vs. J for $\theta < 60^\circ$. C_{Tz} increases with θ at a given J . C_{Tz} vs. J changes at $\theta \sim 60^\circ$, whence C_{Tz} begins to increase with J . C_p remains independent of θ until $J \leq 0.4$. Beyond $J = 0.4$, the freestream momentum starts to become dominant, as the propeller's ability to redirect the freestream momentum at higher θ decreases, which can be seen in the smoke flow visualization in Figure 8. The C_p for the different θ cases start to diverge, and the overall trend is similar to that of C_{Tz} . Again, the overall trend for C_p vs. J changes at $\theta \sim 60^\circ$, where the value of C_p increases with J , which is also seen in the results from [16].

As suggested in [15-16], data collapses better using a normalized advanced ratio, J_{z0} :

$$J_{z0} = \frac{V \cos \theta}{n D} \quad (2)$$

Figure 9 replots the results in Figure 7 vs. J_{z0} . Results for both coefficients collapse at $\theta \leq 60^\circ$. At higher θ , the curves do not collapse as well, and the approach becomes meaningless at $\theta = 90^\circ$.

The propeller efficiency, η , can also be renormalized using J_{z0} :

$$\eta_z = \frac{C_{Tz} J_{z0}}{C_p} \quad (3)$$

The result is plotted in Figure 10. The efficiency for all θ cases collapses, except for $\theta = 90^\circ$ where the value of J_z is equal to zero, which results in $\eta_z = 0$. The collapse in Figure 10 is presumably also evidence of Reynolds number independency.

The collapse of the C_T , C_p and η_z renders simple the prediction of performance at $\theta < 60^\circ$ using the data at $\theta = 0^\circ$, because from Figure 9, it is evident that

$$C_T(J_{z0})_\theta = C_T(J)_{0^\circ} \text{ and } C_P(J_{z0})_\theta = C_P(J)_{0^\circ} \quad (4)$$

Where $\theta < 60^\circ$. This collapse of data between 0° and 60° was used to predict performance of the 14 x 14 propeller (Figure 10) at $\theta = 45^\circ$ and 83.3 RPS. Since the 14x14 propeller is has a high pitch to diameter ratio, the changes in C_{Tz} at $\theta < 60^\circ$ and $J < 0.4$ are relatively low. C_{Tz} declines with increasing J at $J \geq 0.4$, which agrees with the overall trend seen in Figure 7.

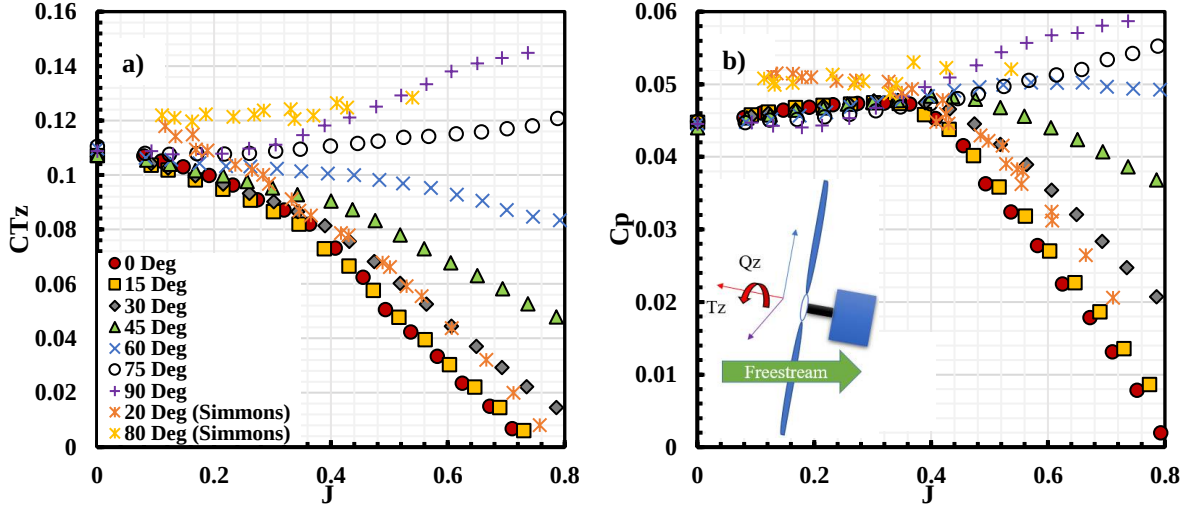


Figure 7. a) C_{Tz} and b) C_p for 11x7 propeller vs. J and θ (100 RPS)

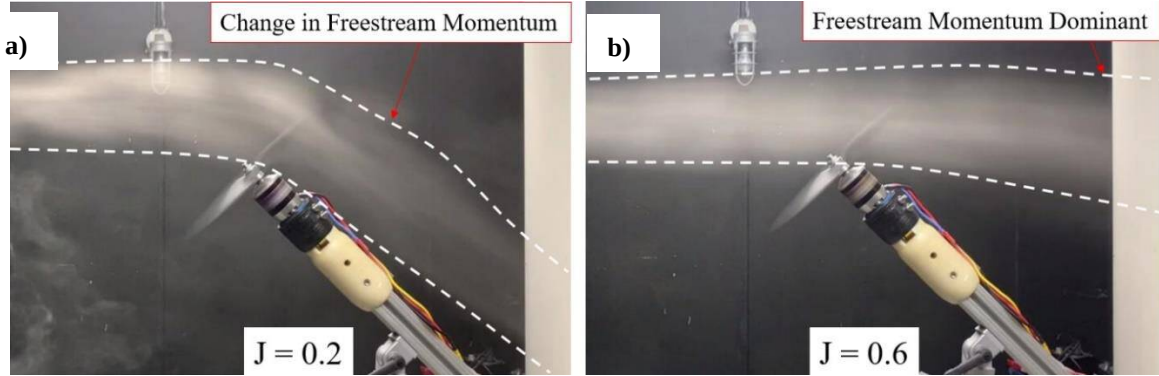


Figure 8. Smoke visualization for APC 11x7 propeller at $\theta = 45^\circ$ under a) $J = 0.2$ and b) $J = 0.6$

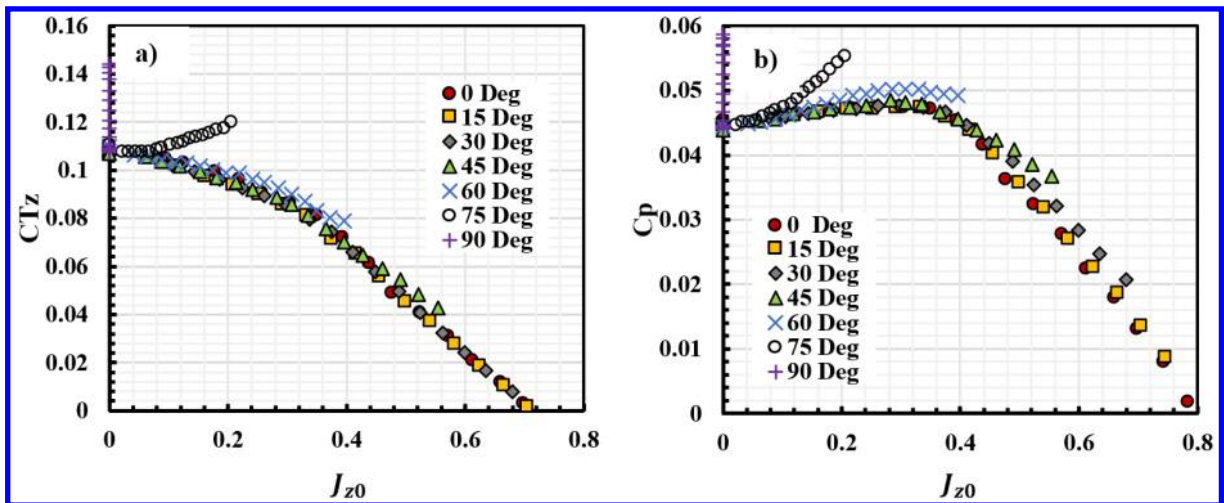


Figure 9. Replotting a) C_{Tz} and b) C_p for 11x7 propeller v.s J_{z0} at different incidence angle.

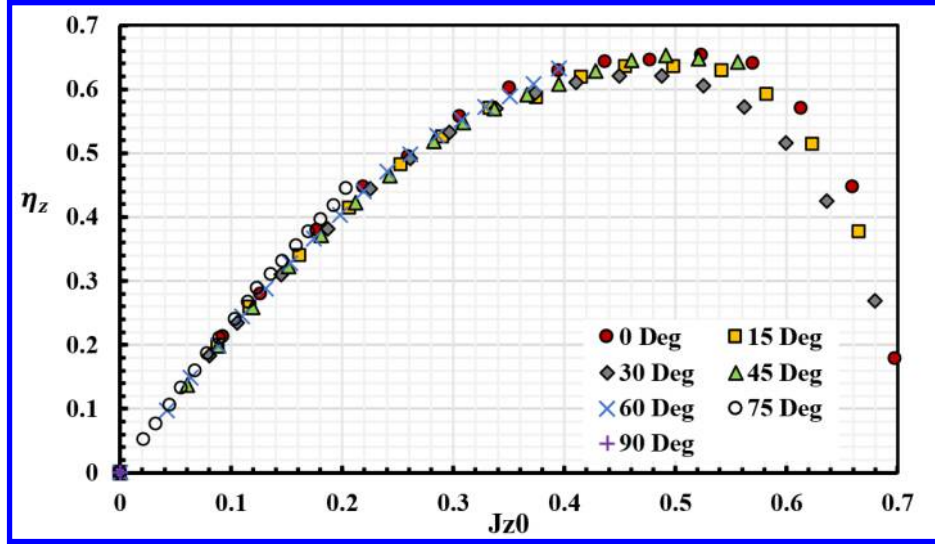


Figure 10. Efficiency for 11x7 propeller calculated based on normalized advance ratio J_z

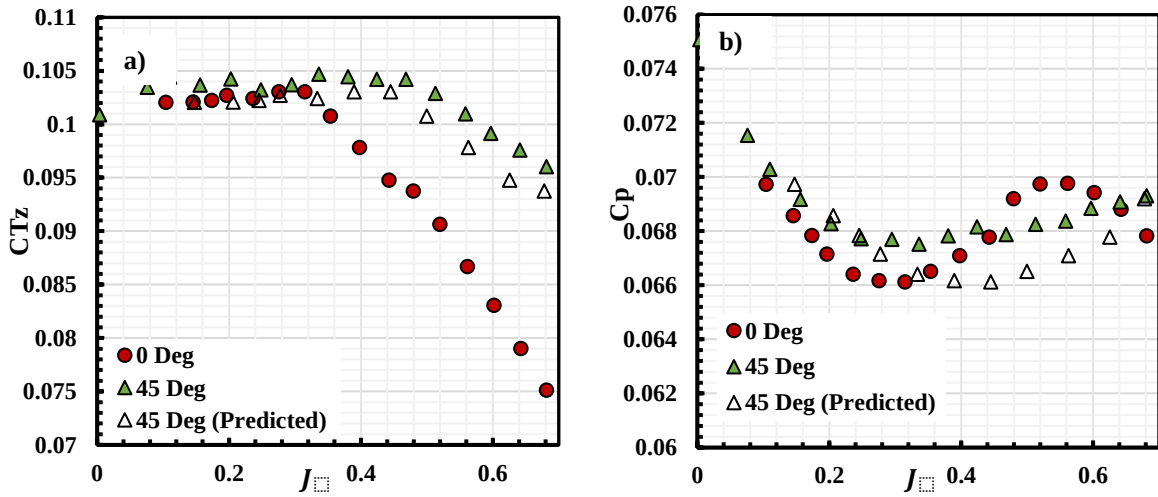


Figure 11. Measurement vs. prediction of a) C_{Tz} and b) C_p for 14x14 propeller vs. J and θ (83.3 RPS)

For C_p , at small θ and $J < 0.4$, a reduction is observed with the increment of J , possibly due to the blade stall. C_p then remains relatively constant at higher J . For large θ , trends resemble those in Figure 7. Both predicted coefficients for $\theta = 45^\circ$ follow closely the respective experimental results.

The propeller pitching moment coefficient, C_{Qx} , and rolling moment coefficient, C_{Qy} , vs. J at different incidence angles are shown in Figure 12. A positive C_{Qx} is pitch-down, while a positive C_{Qy} is counterclockwise. Unlike the result in [10], a clear trend is seen for both coefficients. C_{Qx} decreases while C_{Qy} increases with J for all $\theta > 0$. Both phenomena are a result of asymmetric blade loading due to interaction with the free stream. As the APC propeller tested is not foldable and the pitch is fixed, and natural frequency for the first bending structural mode (about 231 Hz for the 11-inch propeller [19]) is much higher than the operational RPS in all cases, flapping motion is not significant. Thus, the advancing blade will encounter a higher local velocity than the retrieving blade, resulting in a higher blade sectional lift and drag. With increase of either advance ratio or incidence angle, the phenomena will be amplified, resulting in the trend seen in Figure 12. Also,

a noticeable reduction in both C_{Qx} and C_{Qy} curve slope occurs at $0 < \theta < 45^\circ$ at $J \sim 0.4$. Again, this is the threshold where the freestream momentum starts to dominate as discussed previously.

An alternative method to normalize the advance ratio at different incidence angle follows classical helicopter ideas [1,15-16]:

$$J_{z90} = \frac{V \sin \theta}{n D} \quad (5)$$

Using J_{z90} , Figure 12 is replotted in Figure 13. C_{Qx} nearly collapses onto one universal line, C_{Qy} follows one line before peeling off at discrete values of advance ratio for different respective values of θ . As the helicopter-inspired model is constructed to better account for full edgewise flight (90 degrees in our nomenclature), it is not surprising that collapse to a common line for the highest incidence angles, is better than for 15 degrees to 45 degrees. Such linear trend provides a similar prediction method for the propeller pitching moment at different θ as Equation 5:

$$C_{Qx}(J_{z0})_\theta = C_{Qx}(J)_{90^\circ} \text{ and } C_{Qy}(J_{z0})_\theta = C_{qy}(J)_{90^\circ} \quad (6)$$

The measured and predicted C_{Qx} and C_{Qy} for the 14x14 propeller are shown in Figure 14, where the trend follows that of Figure 12. Again, the predicted C_{Qx} and C_{Qy} using Equation 6 for $\theta = 45^\circ$ case follows closely to the measured experimental results.

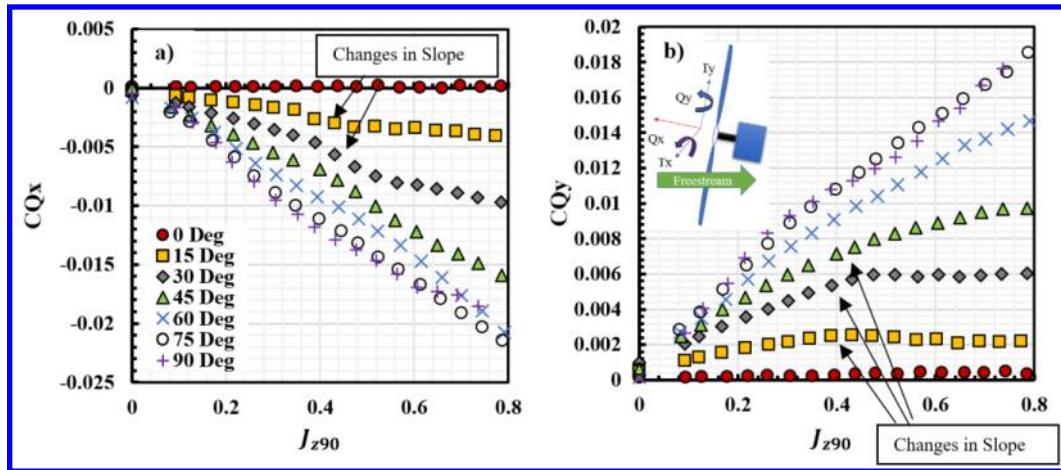


Figure 12. a) C_{Qx} and b) C_{Qy} for 11x7 propeller vs. J at different incidence angles

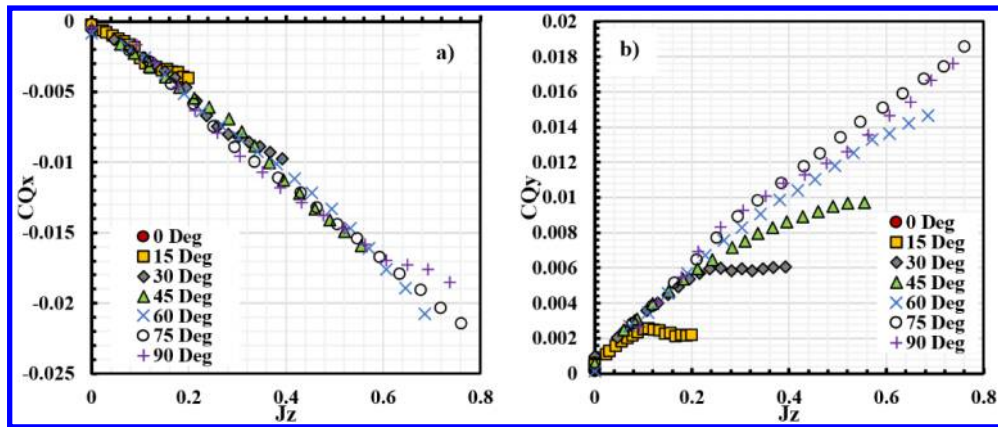


Figure 13. Replotting a) C_{Qx} and b) C_{Qy} for 11x7 propeller vs. J_{z90} at different incidence angles

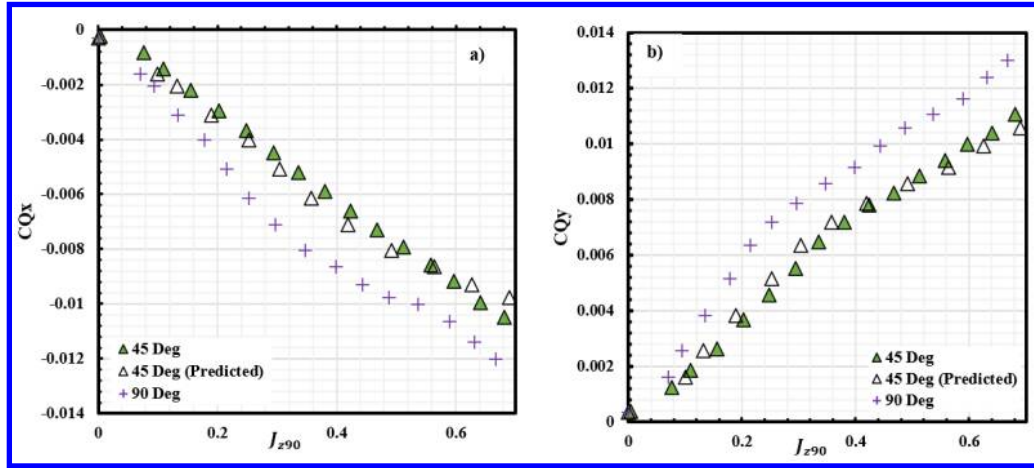


Figure 14. Measurement vs. prediction of a) C_{Tz} and b) C_p for 14x14 propeller vs. J and θ (83.3 RPS)

B) Propeller Performance under Time-Varying Sinusoidal Freestream

The steady freestream data is used to create a model to compare with the time-varying freestream experiments. The force/thrust coefficient on each axis, along with the torque/power coefficient on each axis, are input to a two-dimensional linear interpolation, which accounts for the changes in propeller Reynolds number and advance ratio. The propeller rotational speed and the instantaneous freestream velocity are measured under time-varying freestream conditions, from which the instantaneous advance ratio is calculated. This is then input into the previously developed 2-D linear interpolation between each data point to get the predicted steady-state propeller response, which can be written as:

$$C_{T_{Predicted}} = C_{T_{Steady}}(J_{measured}) \quad (7.1)$$

$$C_{Q_{Predicted}} = C_{Q_{Steady}}(J_{measured}) \quad (7.2)$$

$$C_{P_{Predicted}} = C_{P_{Steady}}(J_{measured}) \quad (7.3)$$

Because the electronic speed controller running the propeller-motor does not have a governor, motor RPS will change, for a constant throttle input setting, depending on the load encountered. This is captured in eqns. 7. Figure 15 shows the measured and predicted thrust coefficients, $C_{T,m}$ and $C_{T,p}$, and measured and predicted power coefficients, $C_{P,m}$ and $C_{P,p}$ for the APC 11x7 propeller at 100 RPS under $\omega_k = 0.0026$ at $\theta = 0^\circ$. When the freestream is reduced, the instantaneous propeller advance ratio falls, results in an increment in both C_{Tz} and C_{Pzp} , as seen in Figure 7. The steady-state model overlaps the measured value for both thrust and power, implying that the response of the propeller is quasi-steady. That is, the propeller responds to change in advance ratio directly.

The measured and predicted axial thrust and power for the higher γ/D propeller, APC 14x14, at 83.3 RPS and $\omega_k = 0.0033$ at $\theta = 0^\circ$ are shown in Figure 16. The steady-state model provides a good prediction for both thrust and power, despite the higher noise in the data, possibly caused by blade stall. In sum, the quasi-steady fit is effective for both propellers, under all RPS values tested at $0.12 < J < 0.7$ for both $\omega_k = 0.0033$ and 0.0147.

At the highest available shuttering system frequency ($\omega = 2.25 \text{ Hz}$), the quasi-steady model is still applicable for both 11x7 ($\omega_k = 0.0625$) and 14x14 ($\omega_k = 0.0796$) propellers for the thrust coefficient. Figure

17 shows the measured and predicted thrust. The predicted thrust for 11x7 propeller overlaps on the measured thrust, followed the trend in Figure 15, albeit with the caveat of noise for the 14x14 propeller data set.

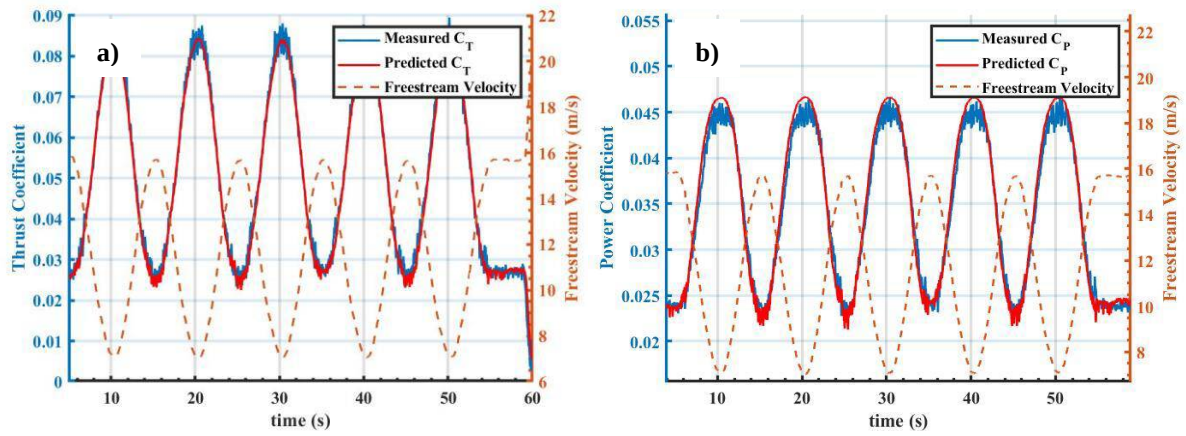


Figure 15. Measured and predicted a) C_T and b) C_P at $\omega_k = 0.0026$ for APC 11x7 propeller at 100 RPS ($\theta = 0^\circ$)

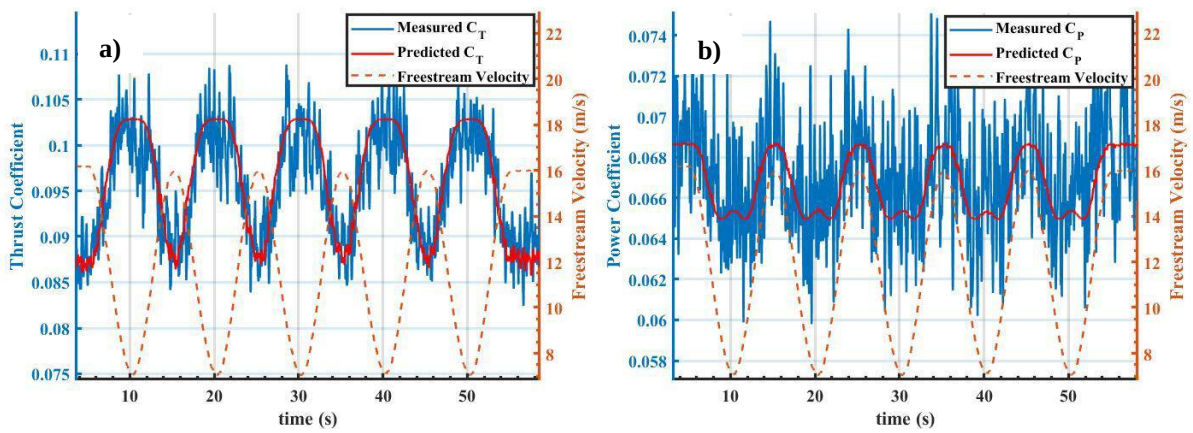


Figure 16 Measured and predicted a) C_T and b) C_P at $\omega_k = 0.0033$ for APC 14x14 propeller at 83.3 RPS ($\theta = 0^\circ$)

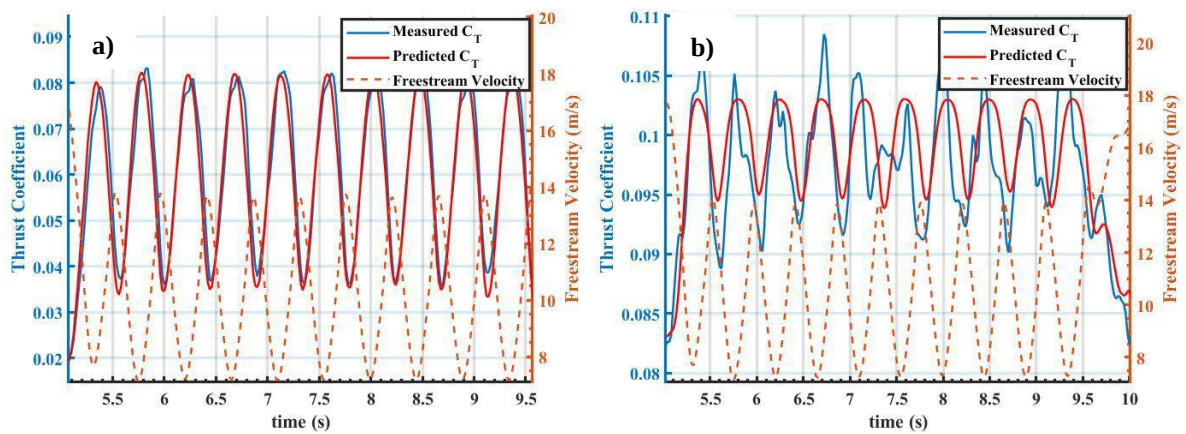


Figure 17 Measured and predicted C_T for a) 11x7 at 100 RPS and $\omega_k = 0.0625$ b) 14x14 propeller at 83.3 RPS and $\omega_k = 0.0796$ ($\theta = 0^\circ$)

However, the power coefficient does not fit the quasi-steady model as well. The power coefficient for the 11x7 propeller at $\omega_k = 0.0515$ and $\omega_k = 0.0625$ is shown below in Figure 18. In particular for the highest normalized frequency, the quasi-steady fit overpredicts the magnitude of the variation in measured power coefficient.

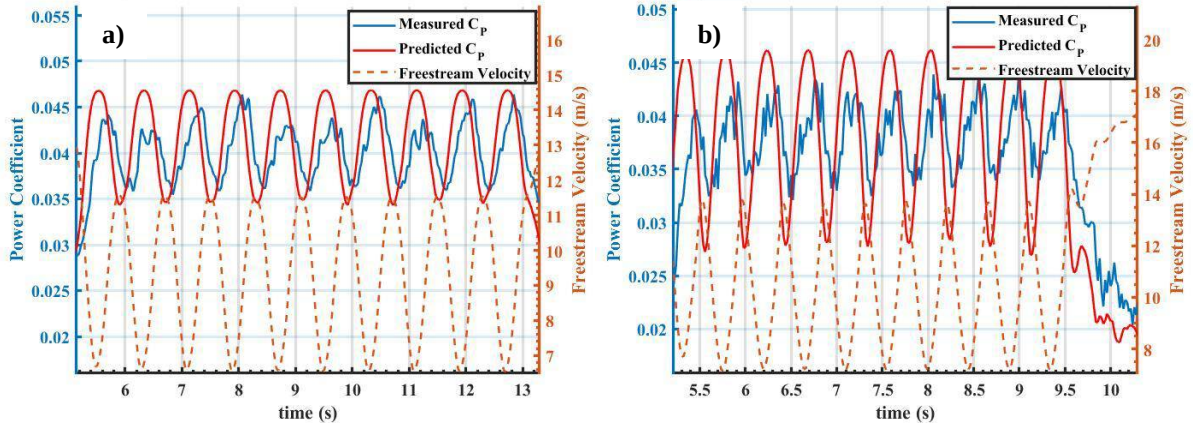


Figure 18 Measured and predicted power for 11x7 at 100 RPS with a) $\omega_k = 0.0515$ and b) $\omega_k = 0.0625$ ($\theta = 0^\circ$)

Figure 19 shows the measured and predicted axial thrust and power for the 11x7 propeller at $\theta = 90^\circ$ at $\omega_k = 0.0625$. The quasi-steady model again provides good prediction of the propeller axial thrust, doing somewhat worse in the power coefficient prediction.

At higher propeller incidence angle, the changes in propeller disk pitching and rolling moment coefficient become important. Figure 20 shows the measured Q_x and Q_y for the 11x7 propeller at $\omega_k = 0.0625$ and 100 RPS, along with the quasi-steady prediction, again with a good fit.

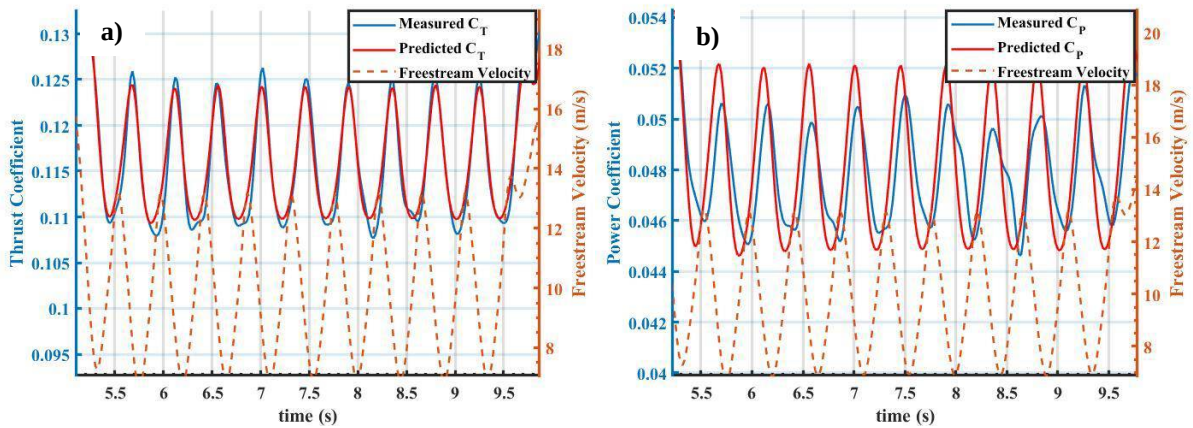


Figure 19. Measured and predicted a) C_T and b) C_p for 11x7 with $\omega_k = 0.0625$ at 100 RPS ($\theta = 90^\circ$)

Experimental results for the intermediate ω and θ cases for the APC 11x7 propeller at 100 RPS are shown in Figure 21 in the compact form of C_{Tz} vs. C_p . The measurements are in blue, and the mean of the measurement is in red; that is, for each value of thrust, the data-points for power, are averaged. The corresponding steady curve of C_{Tz} vs. C_p (Figure 7) is in black.

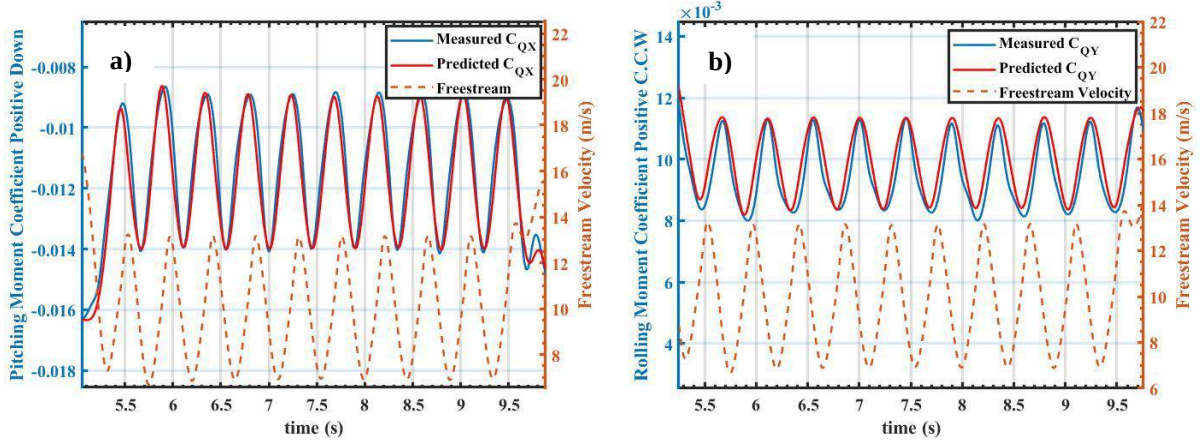


Figure 20. Measured and predicted propeller pitching moment C_{Qx} and b) C_{Qy} at $\omega_k = 0.0625$ for APC 11x7 Propeller at 100 RPS ($\theta = 90^\circ$)

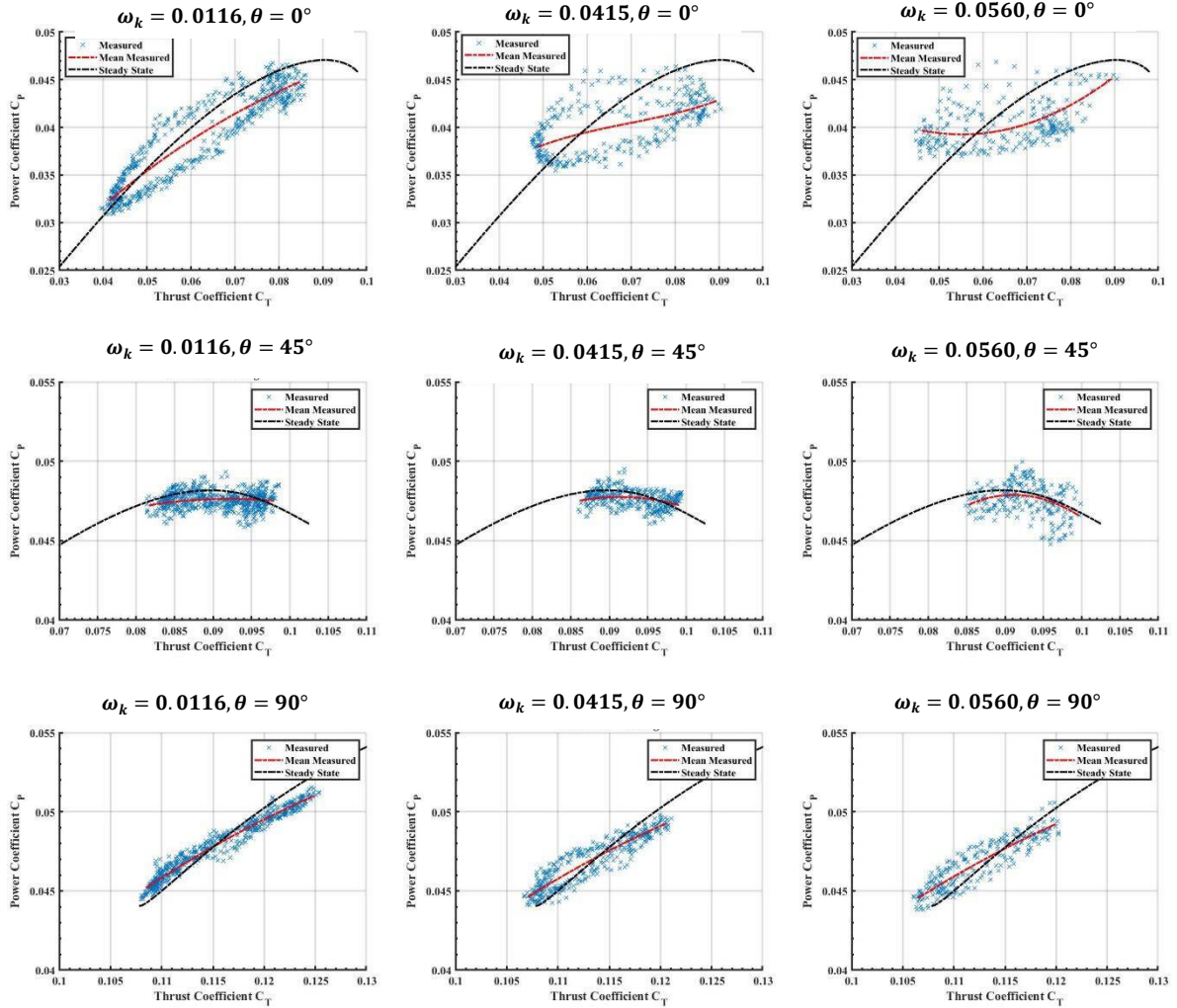


Figure 21. C_{Tz} v.s C_P for APC 11x7 propeller with $\omega_k = 0.0116, 0.0415$ and 0.0560 at $\theta = 0^\circ, 45^\circ$ and 90° at 100 RPS.

For $\theta = 0^\circ$ the scatter is larger than $\theta = 45^\circ$, which in turn is larger than for $\theta = 90^\circ$. Agreement between model and measurement also improves as incidence angle increases. The implication is that at least for

the shuttering frequency range realizable in this experiment, full edgewise flight has less unsteady dynamic interaction between the time-varying free stream and the propeller integrated coefficients. Interestingly, and perhaps oddly, the most complicated case is that of direct head-on flight (in the sense of conventional fixed-wing-aircraft), with that of drone at large incidence angle being an intermediate case.

V. Conclusions

All six components of forces and torques were measured in an open-jet wind tunnel for a pair of propellers: moderate ratio of pitch to diameter, and high. The latter case is expected to stall at low advance ratio, and this was manifested in higher data-scatter. Propellers were oriented conventionally, facing the flow; in edgewise flight; and in 15-degree incidence-angle increments in between, the latter cases of relevance to multi-copter drones in forward flight. The test section speed was both temporally steady and sinusoidally varied. An increase in thrust, power, pitching moment, and rolling moment was found with the increment of incidence angle, consistent with the helicopter literature. Plotting the thrust and power coefficient with respect to the normalized advance ratio, J_z , accounting for propeller orientation with respect to the flow, results in collapse of the various incidence-angle cases to a common curve, for all but entirely edgewise flight. In the latter, an alternative normalization is commensurately successful.

The steady free-stream data were used to build a quasi-steady model to compare with measurements of coefficients of thrust, power, and of rolling-moment and pitching-moment with respect to the propeller axis, for reduced frequencies (based on free stream and propeller diameter) of up to 0.0796. Good fit between model and measurement, for all up to the highest realized frequencies, and especially in purely edgewise flight, shows quasi-steady response. This behavior is expected to break down as reduced-frequency increases, but that is beyond the present range of the apparatus.

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